

FROM CLINICAL GUIDELINES TO COMPUTABLE DECISION LOGIC FOR NEUROSymbOLIC AI

DATASCI290: NEUROSymbOLIC AI, SPRING 2026

MARCH 4 2026

REFERENCES

- Kandula, Vasudhan Varma, and Pushpak Bhattacharyya. *"Decision knowledge graphs: Construction of and usage in question answering for clinical practice guidelines."* arXiv preprint arXiv:2308.02984 (2023).
- Dwyer, Owen P., Teodoro Baldazzi, Jim Davies, Emanuel Sallinger, and Adriano Vlad. *"Reasoning over health records with Vadalog: a rule-based approach to patient pathways."* CEUR Workshop Proceedings, 2023.

CLINICAL PRACTICE GUIDELINES

- **Clinical Practice Guidelines (CPGs)** are *evidence-informed recommendations* that standardize clinical decision-making by specifying
 - **who** a recommendation applies to
 - under **what conditions**,
 - **which** exclusions/red flags override
 - **what** action to take
- **Intended use settings**
 - **Point-of-care decisions**: clinician decision support during diagnosis, workup, treatment selection, follow-up planning.
 - **Care pathways / protocols**: hospitals and clinics translating best practices into consistent workflows.
 - **Quality & safety**: reducing unwarranted variation; supporting audits, quality measures, and guideline adherence checks.
 - **Training and reference**: clinical education and “what should we do next?” reference for trainees and staff.
 - **Patient-facing triage (adapted guidance)**: after-hours symptom guidance and escalation rules (often derived/simplified from clinician CPG logic).

"Patient can be treated with chemotherapy if age less than 65 and greater than 35. They should not have any substantial comorbidities."

CPG EXAMPLES

Infants younger than three months with a rectal temperature of **38.0°C (100.4°F) or higher** should be **evaluated by a clinician urgently**, even if they otherwise appear well. Caregivers should seek immediate medical assessment rather than monitoring the child at home.

"Patient can be treated with chemotherapy if age less than 65 and greater than 35. They should not have any substantial comorbidities."

Patients whose asthma symptoms **worsen despite use of a quick-relief inhaler**, or who experience **difficulty speaking full sentences, persistent wheezing, or increasing breathlessness**, should **seek urgent medical care immediately**. These symptoms may indicate a severe asthma exacerbation requiring prompt evaluation and treatment.

CHECK SOME CPGS OUT

Seattle Children's
HOSPITAL • RESEARCH • FOUNDATION

I want to... Search

Your Visit Conditions Clinics Research **Health & Safety**

Health & Safety > Prevent and Treat Illness >

Health & Safety

Prevent and Treat Illness

Fever



What is fever?

It is normal for the body's temperature to change during the day. Exercise, warm clothing and hot weather can cause a small rise in temperature. A fever is a body temperature of over 100.4 degrees F (or 38.0 degrees C). Most of the time fever is not harmful. Fever is the body's way of fighting an illness.

Treatment of a Fever

You do not always need to treat a fever. The best thing to do is to watch how your child acts. Are they eating and sleeping well? How do they look? Do they play at times? If they are not comfortable, you can try these tips.

Things you should do for a fever:

- Offer extra fluids to drink all through the day.
- Dress your child in comfortable clothes or pajamas. Avoid too many clothes or bundling. This can raise body temperature.
- Decide if your child is comfortable. If not, you can give **acetaminophen** (Tylenol) if your child is over 3 months old. You can give **acetaminophen** or **ibuprofen** (Motrin) if your child is over 6 months old. Make sure to give the correct dose based on your child's weight. Use the measuring tool that came with the medicine. Follow the instructions on the medicine.
- Make sure your child gets enough rest. Visit [Activities for Children Sick at Home](#) for ideas for things your child can do when they're not sleeping.

Things you should not do for a fever:

- Do not use rubbing alcohol or ice water to cool your child down.
- Don't worry if your child is not eating solid food. Liquids are needed most during a fever.
- Don't wake your child up to give them fever medicine.
- Don't ever give aspirin to your child.
- Don't give fever medicine to a baby under 3 months old, unless told to by a doctor.
- Don't use ibuprofen for babies less than 6 months unless told to by a doctor.
- Don't switch between acetaminophen and ibuprofen unless told to do so by your doctor. Choose one and stick with the dosing schedule to avoid the risk of giving too much.

Call the doctor if your child:

- Is less than 3 months and has a fever. Call right away.
- Is 3 months to 2 years old, has had a fever for more than 24 hours and has no other symptoms.
- Has a fever for more than 3 days.
- Is in pain or complains of pain, especially in the neck, ears or throat.
- Is very cranky or sleepy.
- Has trouble breathing or is breathing fast.
- Has a seizure.
- Has a rash (except chicken pox).
- Has dry lips, tongue or mouth.
- Vomits often or for more than a day.
- Has not wet a diaper or gone pee in 8 hours.
- Has a chronic health condition that weakens their immune system.
- Has a fever of over 104 degrees F (or 40 degrees C).
- Is not alert when awake (lethargic).

iaBPG
INTERNATIONAL
AFFAIRS & BEST PRACTICE
GUIDELINES
TRANSFORMING
NURSING THROUGH
KNOWLEDGE

Clinical Best
Practice Guidelines

MAY 2017

Adult Asthma Care: Promoting Control of Asthma

Second Edition



RNAO
Registered Nurses' Association of Ontario
l'Association des infirmières et infirmiers
associés de l'Ontario

■ <https://www.seattlechildrens.org/health-safety/illness/fever>

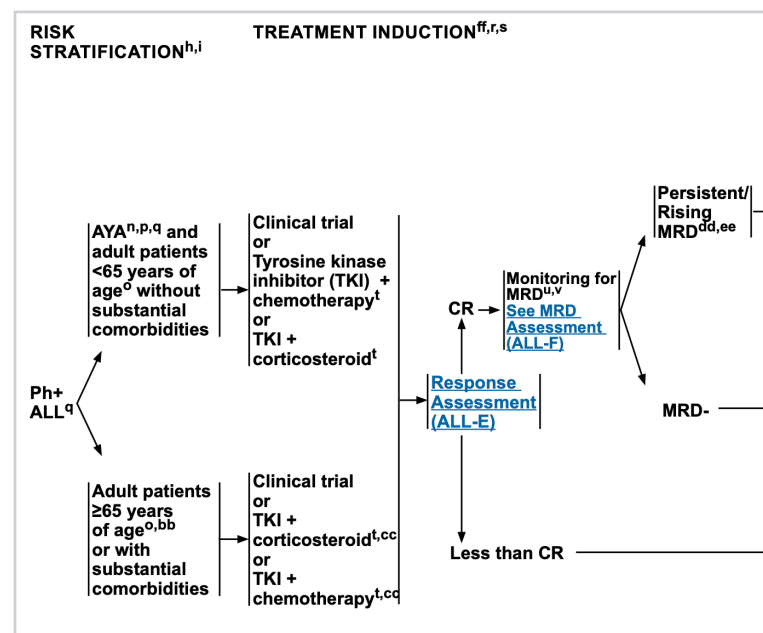
■ <https://rnao.ca/bpg/guidelines/adult-asthma-care>

THE DECISION KNOWLEDGE GRAPH

- The **static** and ... the **dynamic**
- The key design principle in the DKG model: the **separation** between **static medical knowledge** and **dynamic decision constraints**.
 - Static components: include disease concepts, treatments, diagnostic procedures, and clinical terminology hierarchies.
 - Decision components: include eligibility constraints such as patient age, laboratory thresholds, symptom combinations, and contraindications.
- This separation makes the system robust to frequent guideline updates
 - Only decision nodes require modification.

CANCER CPG

- Question-Answering system: *A model trained to answer a question*
- CPGs for National Comprehensive Cancer Network (NCCN)
 - <https://www.nccn.org/guidelines/nccn-guidelines>
- This guideline
 - **Initial treatment depends on patient fitness.** Younger/fit patients (AYA and adults ≤65 without major comorbidities) are typically treated with a clinical trial regimen using a Tyrosine Kinase Inhibitor (TKI) plus corticosteroids, while older or less fit patients may receive TKI combined with chemotherapy or other modified regimens.
 - **Doctors then evaluate whether the leukemia goes into remission (CR).** After initial treatment, they check whether the disease has responded and whether complete remission has been achieved.
 - **If remission occurs**, patients are monitored closely using “MRD testing”.
 - **Future treatment decisions depend on MRD results.** If MRD persists or rises, therapy may need to be intensified or changed; if MRD remains negative, treatment continues along the planned pathway.



QUESTION-ANSWERING OVER PATIENT CONSTRAINTS PLUS CPGS

- Although clinical practice guidelines are central to evidence-based medicine, real-world adherence is variable: studies typically find that only 40–60% of recommended guideline actions are implemented in practice, with non-adherence rates approaching 60% in some settings.
- The main objective of a Decision Knowledge Graph (DKG) is to perform question-answering
 - Thus **reducing the manual effort of a doctor** to search through the guidelines.

What is next treatment advice given a patient's constraints ?

For instance: A patient is ALL positive. After his initial diagnosis he is classified as ph- patient. His age is 65. He is not treated with other cancer treatments. What treatment is recommended in this condition?

Given a patient's medical constraints and treatment stage, whether a particular treatment is advisable or not?

Ex: A patient is ALL positive. After his initial diagnosis he is classified as ph- patient. His age is 65. He is not diagnosed with any other cancer treatment. Can we perform TKI+ Chemotherapy on him?

What are the patient's medical constraints that needs to be satisfied given a treatment stage ?

Ex: A patient is ALL positive. After his initial diagnosis he is classified as ph+ patient. What are patient constraints for doing chemotherapy?

DECISION KNOWLEDGE GRAPH: STRUCTURAL COMPONENTS

- DKG representation: frequent node types:
 - Guideline
 - Recommendation
 - Constraint, Action
 - EvidenceGrade
 - Version
- **Recommendation node**
 - represents a single actionable statement from a clinical guideline.
- **Constraint nodes**
 - encode eligibility criteria such as laboratory thresholds, demographic ranges, or symptom conditions.
- **Action nodes**
 - represent clinical interventions such as ordering a test, prescribing a drug, or referring a patient.

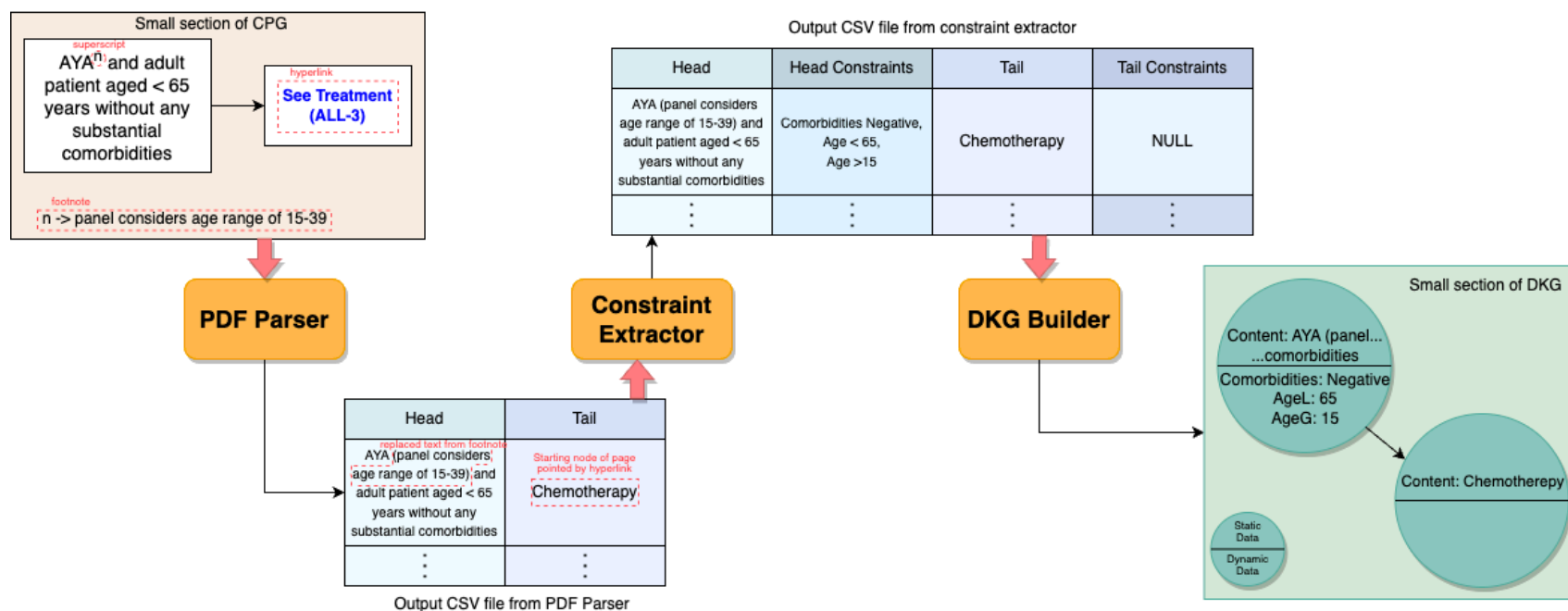
GRAPH RELATIONSHIPS IN THE DKG SCHEMA

- Relationships capture the logical structure of guideline recommendations.
 - HAS_RECOMMENDATION
 - links a guideline document to individual recommendation nodes.
 - HAS_CONDITION
 - connects recommendations to the constraints that must hold for applicability.
 - HAS_EXCLUSION
 - represents contraindications or red flag symptoms.
 - RECOMMENDS_ACTION
 - connects the recommendation node to the resulting clinical intervention.

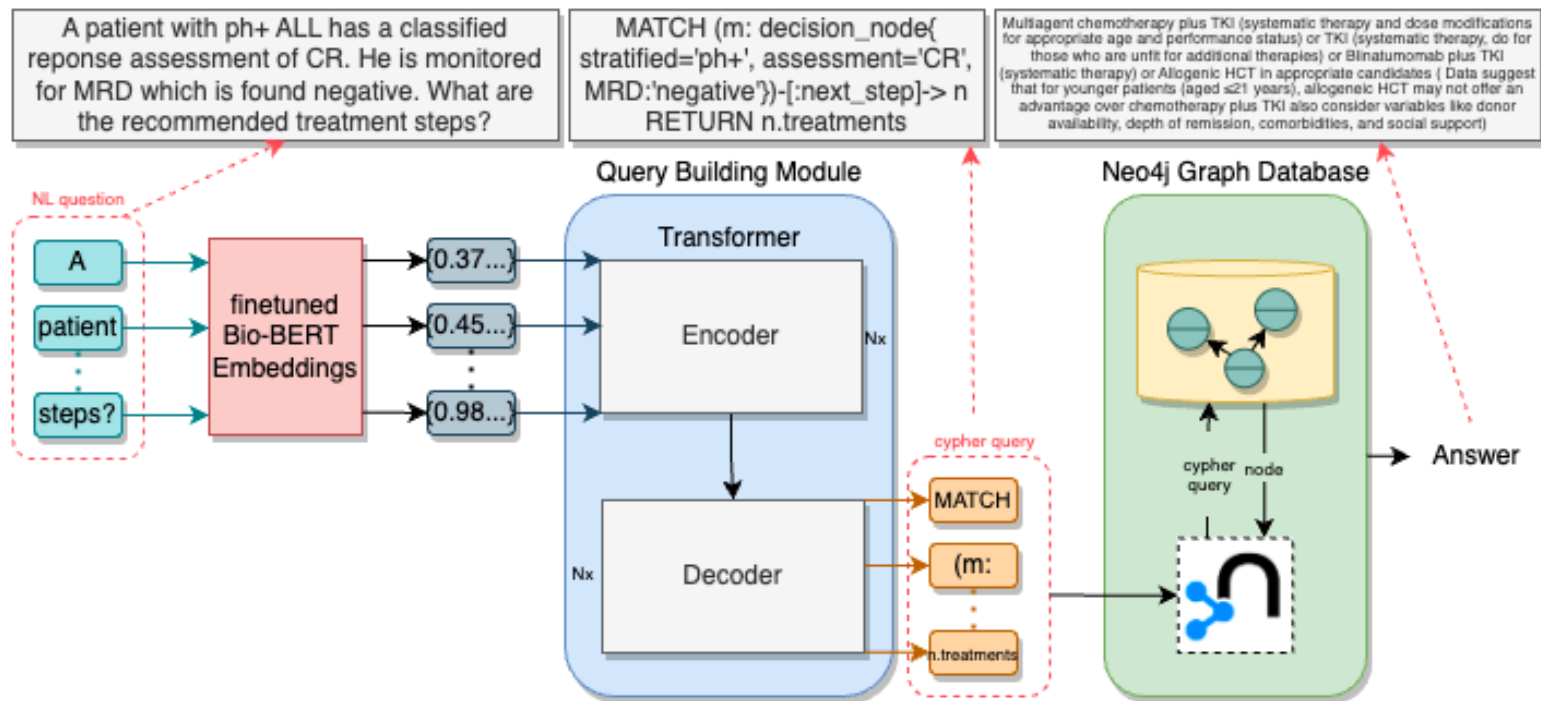
EXAMPLE REPRESENTATION: OF A CLINICAL RECOMMENDATION

- A **recommendation node** representing a **treatment decision** may connect to **multiple constraint nodes** that collectively define eligibility.
- For example, a recommendation might connect to a constraint nodes:
 - 'age between 18 and 65'
 - 'blood pressure above threshold'
 - 'no contraindicated medication'.
- Only when all constraints are satisfied does the recommendation node link to the action node representing the treatment.
 - Done via CYPHER queries

DKG CONSTRUCTION



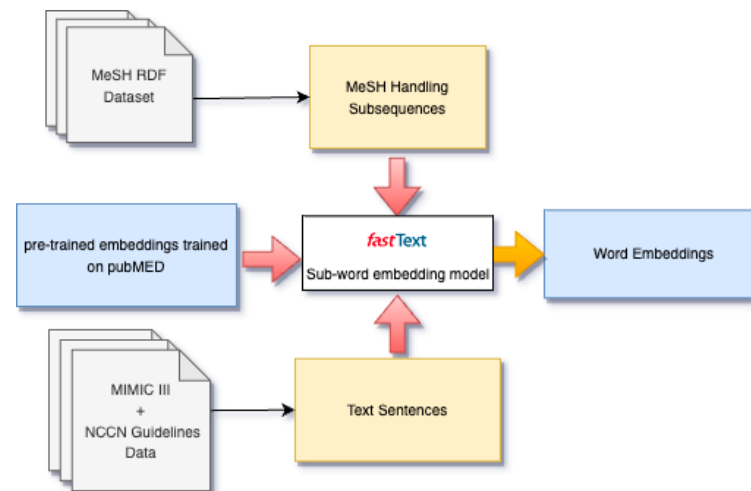
QUESTION ANSWERING USING DKG



■ Question Answering using DKG

- i) Query Building Module builds the cypher query from given natural language (NL) question
- ii) Neo4j Graph Database fetches the node from the DKG according to the query and returns the content of the node

EMBEDDINGS PIPELINE



- Pipeline trains domain-specific biomedical word embeddings using FastText on MeSH, clinical records, and guideline text
 - so that the system can understand semantic relationships between medical terms
- **Moot point now, we have LLMs !**
 - And can use LLM +KG-RAG

EVALUATION

Question: A 68-year-old ph-ALL patient without any significant comorbidities underwent a clinical trial during the treatment induction phase, achieving a CR response assessment. He was monitored with persistent rising MRD. What procedures are recommended?

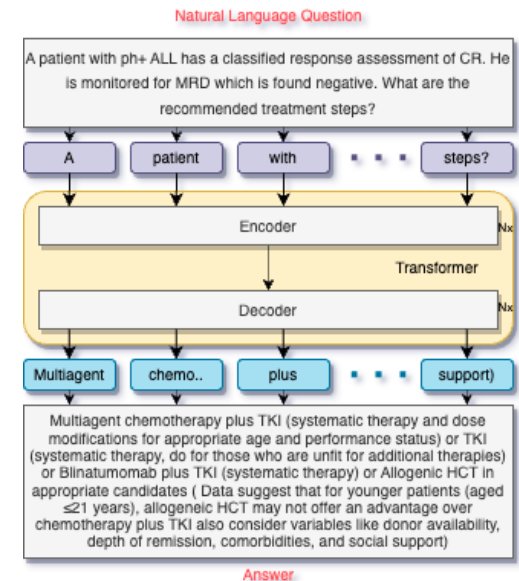
- **Actual Answer:** Blinatumomab followed by Allogenic HCT

- **Predicted Answer (without DKG):** Predicted Answer: Allogenic HCT (especially if high-risk features or consider continuing multiagent chemotherapy or Blinatumomab

- **Predicted cypher query:**

```
MATCH (m: decision_decision_
node stratified='ph-', MRD:'rising')-
[:next_step]-> n RETURN n.treatments
```

- **Predicted Answer (with DKG):** Blinatumomab followed by Allogenic HCT



EVALUATION SETUP

- Evaluate DKG approach using a question answering task over clinical practice guidelines.
- Baseline model uses BioBERT to answer questions **directly from textual guideline documents**.
- The DKG-enhanced system first queries the decision knowledge graph to retrieve relevant decision structures before generating answers.
- The architecture integrates structured knowledge retrieval with language model reasoning.

```
[  
  {  
    "QUESTION": String,  
    "ANSWER": String,  
    "QUERY": String,  
    "Expected_Node": Integer,  
    "DKG_response": Integer,  
  }, ...  
]
```


RESULTS

Metric	Without DKG	With DKG
ROUGE precision	0.49	0.95
ROUGE recall	0.62	0.96
ROUGE f-measure	0.51	0.96
BLEU	0.44	0.95
Jaccard	0.46	0.92
Accuracy	0.259	0.676

- **ROUGE (Recall-Oriented Understudy for Gisting Evaluation)** Family of metrics mostly used in **summarization**.
 - **ROUGE-N**: overlap of **n-grams** (word sequences) between generated text and reference.
 - ROUGE-1 = unigram overlap, ROUGE-2 = bigram overlap.
 - Typically reported as **Precision / Recall / F1**:
 - **Precision** = overlap / # n-grams in generated text
 - **Recall** = overlap / # n-grams in reference text
 - Intuition: “How much of the reference content did you cover?” (especially recall).
- **BLEU (Bilingual Evaluation Understudy)** Originally for **machine translation**, often used for generation too.
 - Measures **modified n-gram precision** (usually 1–4 grams): “How many of my n-grams appear in the reference(s)?”
 - Includes a **brevity penalty** so very short outputs don’t score artificially high.
 - Intuition: rewards **exact phrasing** similar to reference; tends to be harsh on valid paraphrases.
- **Jaccard (Jaccard similarity / Intersection over Union)**
 - A simple set-similarity score.
 - For sets AAA and BBB:
 - $J(A,B) = |A \cap B| / |A \cup B|$
 - In NLP it’s often computed over:
 - **set of words**, or
 - **set of n-grams**, or
 - sometimes **tokens** (implementation varies).
 - Intuition: “What fraction of unique items are shared?”

FAST FORWARD TO TODAY

- **Replace fine-tuned QA models with LLM-based query interfaces**
 - They used BioBERT QA models to answer guideline questions. Today we would simply use an LLM to translate questions into structured queries.
- **Cypher generation does not require model training**
 - Instead of training a neural model for query generation, a prompted LLM can reliably generate Cypher queries over a well-defined schema.
- **Prompting + schema grounding is usually sufficient**
 - With the schema and example queries in the prompt, modern LLMs can produce accurate graph queries without additional model training.



WHERE MIGHT THE (DECISION)
KNOWLEDGE GRAPH GET LIMITED ?



DKG LIMITATIONS (ACKNOWLEDGED)

- The DKG representation captures *guideline structure* effectively
 - But does not fully represent *decision semantics*.
- Complex logical constructs such as nested conditions, temporal constraints, and reusable predicate definitions remain difficult to express using graph properties alone
- Implicitly suggest that rule-based reasoning may be required to complement the graph representation

HOWEVER, WHAT STANDS

- **Structured representation** of clinical guidelines is essential
 - The Decision Knowledge Graph approach correctly models recommendations, eligibility constraints, exclusions, and actions as explicit graph elements rather than buried in narrative text.
- **Separating** static medical knowledge from decision constraints is a good design principle
 - Diseases, drugs, and clinical concepts change slowly, while eligibility criteria and thresholds change frequently. The DKG architecture isolates these layers.
- **Explicit** decision structure improves retrieval and reasoning
 - Representing recommendations and constraints as graph nodes allows precise querying and makes guideline logic inspectable and maintainable.
- Knowledge graphs remain the right backbone for guideline knowledge
 - They support integration with clinical terminologies (SNOMED, ICD, LOINC) and allow structured queries over guideline logic.



MORE COMPLEX RULES/ REASONING



TRANSITION FROM GRAPH REPRESENTATION TO LOGICAL REASONING

- While the graph provides a structural representation of guideline knowledge, decision semantics must still be executed.
- This requires a reasoning engine capable of evaluating eligibility conditions and producing decisions.
- Declarative rule systems such as Datalog provide a natural mechanism for encoding such decision logic

VADALOG REASONING FRAMEWORK

- Vadalogue, a reasoning system based on the “Datalog $\pm$ ” family of logical languages
 - “Warded Datalog”
- Vadalogue is designed to combine rule-based reasoning with knowledge graph data
- It supports recursion, existential reasoning, and efficient rule evaluation over large datasets.
- The system is capable of deriving complex conclusions from structured clinical data.

DATA MODEL

MIMIC-IV

admitted(PatientID, AdmissionID, DateTime)

Admission

Procedure

*procedure(AdmissionID, ProcedureSeq,
ProcedureDate, IcdCode, IcdVer)*

EDB

Event

PrimaryEvent

ColonEvent

next()

IDB

OMOP CDM

dirProcSite(P, S)

Concept

concept(ConceptID, Code)

isA(A, B)

EDB

EventCategory

BUILDING TIMELINES

- MIMIC-IV dataset:freely accessible set of electronic health records
 - (From Beth Israel Deaconess Medical Center, MA)
 - <https://physionet.org/content/mimiciv/3.1/>

```
                                admitted(PatientID, AdmissionID, DateTime),
                                procedure(AdmissionID, ProcedureSeq, ProcedureDate, IcdCode, IcdVer)
→ event(PatientID, AdmissionID, ProcedureDate, ProcedureSeq, IcdCode, IcdVer)

                                event(PatientID, AdmissionIDA, DateA, SeqA, IcdCodeA, IcdVerA),
                                event(PatientID, AdmissionIDB, DateB, SeqB, IcdCodeB, IcdVerB),
                                MinSeq = min(SeqB).
→ minSequence(PatientID, AdmissionIDA, DateA, SeqA, IcdCodeA, IcdVerA, MinSeq)

                                minSequence(PatientID, AdmissionID, Date, Seq, IcdCode, IcdVer, MinSeq),
                                Seq = MinSeq
→ primaryEvent(PatientID, AdmissionID, Date, IcdCode, IcdVer)
```

COHORT RETRIEVAL

■ Cohorts such as:

- “diagnosis → scan → surgery → scan”
- “diagnosis → scan → chemoradiotherapy → radical resection → chemo(radio)therapy → scan”

Path	Count
Surgery.	329
Chemotherapy.	18
SurgerySurgery.	18
Colonoscopy.	7
ChemotherapyChemotherapy.	5
[15 more]	<5

```
path(PatientID, EventIDY, EventIDZ, String, Length),  
next(PatientID, EventIDX, EventIDY, _, _, DescX, DescY),  
NewString = concat(DescX, String), NewLength = Length + 1  
→ path(PatientID, EventIDX, EventIDZ, NewString, NewLength)
```

```
path(PatientID, EventIDX, EventIDY, String, Len),  
firstEvent(PatientID, EventIDX, _),  
path(PatientID, EventIDX, EventIDZ, String', Len'),  
MaxLen = mmax(Length',  
→ pathLength(PatientID, EventIDX, EventIDZ, String, Len, MaxLen),
```

```
pathLength(PatientID, EventIDX, EventIDY, String, Len, MaxLen)  
Len = MaxLen  
→ longestPath(PatientID, EventIDX, EventIDY, String, Len)
```

GUIDELINE CONFLICTS

recommends($G1, D1$), recommends($G2, D2$),
interacts($D1, D2$)
→ conflict($G1, G2, D1, D2$)

recommends($G1, D1$), recommends($G2, D2$),
interacts($D1, D2$)
→ conflict($G1, G2, D1, D2$)

recommends($G1, D1$), $G1 = \text{"Heart failure"}$,
recommends($G2, D2$), potentialComorbidity($G2$)
interacts($D1, D2$)
→ heartFailureConflict($G1, G2, D1, D2$)

heartFailureConflict($G1, G2, D1, D2$),
 $C = \text{mcount}(G2)$
→ counts($G2, C$)

G_2 guideline	C conflicts
Atrial fibrillation	51
Depression in adults	45
Acute coronary syndromes	38
Chronic heart failure in adults	38
Dementia	30
Hypertension in adults	27
Chronic obstructive pulmonary disease in over 16s	26
Type 2 diabetes in adults	21
Chronic kidney disease	16
Neuropathic pain in adults	8
Rheumatoid arthritis in adults	3

HANDLING INCOMPLETE CLINICAL DATA

- Clinical records, as well as real-world diagnosis trajectories frequently contain missing or incomplete information
 - *You have “some” infection*
- Vatalog/Datalog $\pm$ allows *existential variables* to represent unknown entities.
- For example, a rule may infer that a clinical episode exists even if its exact boundaries are unknown.
- Such reasoning enables robust inference in incomplete datasets.

EXPLAINABILITY OF RULE-BASED SYSTEMS

- Rule-based reasoning systems produce explicit derivation traces.
- Each decision can be traced back to the rules and facts that produced it.
- This transparency is critical for clinical auditing and regulatory compliance.
- It also enables clinicians to verify the correctness of system decisions.

SUMMARY AND TAKEAWAYS

- **Clinical knowledge separates into two layers:**
 - *medical knowledge* (diseases, drugs, procedures, ontologies) and
 - *decision knowledge* (guideline conditions, thresholds, exclusions, and recommended actions).
- **Knowledge graphs provide the structural backbone**
 - Organize biomedical concepts and relationships and can also store structured guideline elements (as in the **DKG** approach).
- **Logical rules operationalize decision knowledge.**
 - Declarative rule systems (as in **Vadalog**) enable reasoning tasks such as timeline construction, cohort retrieval, and detection of conflicting guideline recommendations.
- **Neural methods support interpretation and candidate generation**
 - LLMs, embeddings, and graph-RAG techniques can interpret patient narratives, map symptoms to concepts, and retrieve relevant guideline fragments or candidate diagnoses.
- **Symbolic reasoning enforces clinical constraints**
 - Logical rules apply eligibility criteria, thresholds, temporal ordering, and red-flag overrides to ensure recommendations follow guideline logic.
- **Neurosymbolic architecture:** neural components interpret language and retrieve candidates, while structured knowledge and explicit reasoning enforce the medical decision framework and provide traceable explanations.